Topic: Detection of Skin Cancer Using Deep Learning

1. Topic of Interest

Skin cancer remains a pervasive global health challenge, characterized by the uncontrolled growth of abnormal skin cells. It is generally categorized into melanoma and non-melanoma types (such as Basal Cell Carcinoma and Squamous Cell Carcinoma). While melanoma accounts for a smaller percentage of cases (approximately 1%), it is responsible for the vast majority of skin cancer-related deaths due to its high potential for metastasis [1]. The traditional diagnostic workflow involves visual inspection by dermatologists followed by invasive biopsies, a process that is often time-consuming, subjective, and prone to inter-observer variability. The topic of interest for this study is the application of Deep Learning (DL) algorithms to automate the detection and classification of these skin lesions. Deep Learning, a subset of machine learning inspired by the biological structure of the human brain, has revolutionized medical image analysis [2]. Specifically, the focus lies on utilizing Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) to analyze dermoscopic images high-resolution, magnified images of the skin surface. The core objective of this technological intervention is to provide early, accurate, and non-invasive diagnoses, which are critical for improving patient survival rates [4].

Current State of Knowledge:

The current landscape of skin cancer detection has shifted significantly from manual feature extraction to automated deep feature learning. Recent bibliometric analyses indicate a rapid increase in research output between 2019 and 2024, driven by the availability of public datasets like HAM10000 and ISIC [5]. State-of-the-art approaches currently rely heavily on Transfer Learning, where pre-trained models like ResNet, VGG, and DenseNet are fine-tuned for skin lesion classification. For instance, recent studies have demonstrated that models like DenseNet201 can outperform other architectures by capturing fine-grained features necessary for distinguishing between visually similar lesions like nevi and melanoma.

However, the field is currently grappling with the "computational cost vs. accuracy" trade-off. While deep models like ResNet152 offer high accuracy, they require massive computational resources and high-resolution images, which are not always available in resource-constrained clinical settings. Consequently, a new wave of research is exploring lightweight architectures and Transformers. For example, the Compact Convolutional Transformer (CCT) has recently been adapted to process low-resolution images (32 X 32 pixels) while maintaining accuracies above 97%, drastically reducing training times from hours to seconds.

Furthermore, researchers are addressing the challenge of data scarcity and class imbalance. Innovations in data augmentation including geometric transformations and Generative Adversarial Networks (GANs) are being employed to artificially expand small training datasets, allowing custom-built CNNs to perform effectively even with limited original data points[9]. The current state of knowledge suggests that while heavy CNNs are the standard, the future lies in hybrid, efficient models that can operate on diverse image qualities without sacrificing diagnostic precision.

2. Problem Statement

WHO: The primary stakeholders affected are patients suffering from undiagnosed skin lesions and dermatologists who face heavy diagnostic workloads. This issue is particularly acute for populations in developing countries or remote areas where access to specialized dermatological care is scarce.

WHAT: The core problem is the high rate of misdiagnosis and late detection of malignant skin cancers. Visual similarities between benign nevi and malignant melanomas, or between Basal Cell Carcinoma (BCC) and Squamous Cell Carcinoma (SCC), make accurate classification difficult even for experts. Furthermore, existing automated solutions often rely on computationally expensive models that are difficult to deploy on standard hardware.

WHEN: Early detection is time-critical. Melanoma detected in its initial stages is highly curable; however, once metastasis occurs, survival rates drop precipitously. The problem is ongoing, with projections indicating a substantial rise in melanoma cases and related deaths by 2040.

WHERE: This is a global issue, but the gap in diagnostic capability is most profound in regions lacking advanced medical imaging infrastructure. Standard datasets often feature light-skinned populations, potentially biasing models against diverse skin tones found globally.

WHY: Traditional biopsy methods are invasive, painful, and slow. Manual screening is subjective; fatigue or lack of experience can lead to errors. While Deep Learning offers a solution, many existing models suffer from the "black box" problem, high resource consumption, and poor generalization due to unbalanced datasets (e.g., thousands of Nevi images but few Dermatofibroma images).

HOW: Currently, researchers address these issues using Transfer Learning and Data Augmentation. However, a significant gap remains standard deep learning models (like VGG19 or ResNet152) are often too heavy for real-time application and struggle when high-resolution imagery is unavailable. There is a critical need for efficient algorithms that can maintain high sensitivity (recall) to ensure no cancer cases are missed, while remaining lightweight enough for widespread deployment.

Leading to the Research Question: The persistence of these challenges specifically the computational burden and the need for high accuracy with limited or low-quality data necessitates an investigation into optimizing deep learning architectures. This leads directly to the primary research inquiry regarding the efficacy of lightweight versus heavy models in skin cancer detection.

3. Primary Research Question

To what extent can optimized, lightweight deep learning architectures (such as Compact Convolutional Transformers or Custom CNNs) achieve diagnostic accuracy comparable to heavy transfer learning models (like DenseNet or ResNet) when applied to resource-constrained or limited skin cancer datasets?

4. Specific Objectives/Aims

1. To evaluate the classification performance (accuracy, sensitivity, and specificity) of a lightweight Custom CNN and Transformer-based models against traditional transfer learning architectures (e.g., YOLOv7, VGG16) on standard dermatological datasets.
2. To investigate the impact of image resolution reduction and data augmentation techniques on the training time and computational efficiency of deep learning models for skin lesion diagnosis.

5. Hypothesis:

It is predicted that a modified, lightweight Deep Learning architecture utilizing attention mechanisms or optimized convolutional layers will achieve classification accuracy greater than 95% on low-resolution skin lesion images, while reducing the training time by at least 80% compared to standard pre-trained transfer learning models.

6. Literature Review

Overview of Deep Learning in Dermatology

The application of Deep Learning (DL) in dermatology has evolved rapidly over the last decade, transitioning from experimental prototypes to robust diagnostic tools capable of rivaling human expertise. Skin cancer detection, particularly the differentiation between malignant melanoma and benign nevi, relies heavily on the ability of algorithms to discern subtle visual patterns that are often imperceptible to the naked eye. A systematic review conducted by Dildar et al. (2021) establishes the foundational landscape of this field, analyzing research trends from 2011 to 2021. Their work highlights that while various architectures such as Artificial Neural Networks (ANN), K-Nearest Neighbors (KNN), and Generative Adversarial Networks (GANs) have been employed, Convolutional Neural Networks (CNNs) have consistently demonstrated superior performance in image classification tasks due to their ability to automatically extract hierarchical features from dermoscopic images [5].

However, Dildar et al. (2021) also identified persistent challenges that have historically plagued the field, most notably the high computational cost required to train deep networks and the issue of class imbalance in standard datasets. In real-world clinical data, benign lesions vastly outnumber malignant ones, causing models to develop a bias toward the majority class. To mitigate this, they noted the increasing utility of GANs for data augmentation generating synthetic yet realistic lesion images to balance training sets a strategy that has become a staple in modern research pipelines [5].

Global Research Trends and Shifts in Methodology

Building on the foundational knowledge established by Dildar et al., a more recent bibliometric analysis by Supriyanto et al. (2025) maps the trajectory of skin cancer research from 2019 to 2024. Their statistical review of 1,697 articles reveals a significant surge in academic output, driven largely by the public availability of standardized datasets such as HAM10000 and the International Skin Imaging Collaboration (ISIC) archives. Interestingly, their analysis highlights a geographical shift in research contribution, with developing nations like India, Pakistan, and China becoming dominant contributors. This trend is likely driven by the acute shortage of specialized dermatologists in these regions, necessitating the development of automated, AI-driven diagnostic support systems [2].

Supriyanto et al. (2025) further observe a critical evolution in the technical focus of recent studies. While early research focused primarily on simple binary classification (benign vs. malignant), the field is currently pivoting toward more complex tasks such as "Semantic Segmentation" and the integration of "Attention Mechanisms." This shift suggests that the academic community is no longer satisfied with models that merely output a prediction; there is a growing demand for architectures that can localize the lesion precisely and explain which features (e.g., asymmetry or border irregularity) influenced the diagnosis, thereby addressing the "black box" problem inherent in deep learning [2].

Transfer Learning and the Debate on Model Complexity

A dominant strategy in current literature is Transfer Learning the practice of adapting heavy, pre-trained models (originally trained on general object datasets like ImageNet) for medical tasks. Batool et al. (2024) provide a rigorous evaluation of this approach by comparing two sophisticated architectures, InceptionResNetV2 and DenseNet201, on the HAM10000 dataset. Their study addressed the complex challenge of multi-class classification, distinguishing between seven distinct types of skin lesions rather than a simple binary set.

The findings of Batool et al. (2024) offer critical insights into model selection. They demonstrated that DenseNet201, with its densely connected layers that maximize information flow, achieved a superior testing accuracy of 97.4% and an exceptional sensitivity of 98.8%. This high sensitivity is clinically paramount, as it minimizes false negatives ensuring that malignant cases are not missed. Furthermore, their research yielded a counter-intuitive finding regarding classifier design: a "simple" classification model outperformed a complex "hierarchical" model (which attempted to classify benign/malignant first, then the specific subtype). This suggests that in the context of imbalanced medical datasets, adding unnecessary structural complexity to the classification pipeline can sometimes degrade performance rather than enhance it [4].

Addressing Data Scarcity with Lightweight Architectures

While heavy transfer learning models like DenseNet201 offer high accuracy, they are computationally expensive and prone to overfitting when training data is scarce. Tchema et al. (2025) challenged the paradigm that "bigger is always better" by investigating the performance of lightweight models on limited datasets. Their study compared YOLOv7, a state-of-the-art object detection model with 36.9 million parameters, against a custom-built, shallow CNN with only 1.6 million parameters. Using a small, constrained dataset of only 574 original images (augmented to approximately 2,800), they found that their custom, lightweight CNN actually achieved higher accuracy (90.12%) than the complex YOLOv7 model (89.65%) [1].

This finding by Tchema et al. (2025) is significant for resource-constrained environments. It demonstrates that when data is limited, complex models may "memorize" the training data (overfitting) rather than learning generalizable features. In contrast, a well-designed shallow network can provide robust performance with a fraction of the computational cost. Their work emphasizes the importance of data augmentation techniques such as rotation, shearing, and flipping—which allowed their small dataset to effectively train a neural network without suffering from severe overfitting [1].

The Frontier of Efficiency: Transformers and Low-Resolution Imaging

The most radical departure from traditional methodologies is presented by Mahmud et al. (2024), who introduced the "SkinNet-14" framework. While previous studies [4, 5] emphasized the need for high-resolution dermoscopy images to capture fine textures, Mahmud et al. hypothesized that modern Transformer-based architectures could discern malignancy even from low-quality data. They adapted a Compact Convolutional Transformer (CCT) to process images downsized to an extremely low resolution of 32 X 32 pixels [3].

The results of the SkinNet-14 model challenge the long-held assumption that high resolution is strictly necessary for high accuracy. Despite the drastic reduction in image quality, their model achieved a testing accuracy of 97.85% on the HAM10000 dataset, outperforming much heavier models like VGG19 and ResNet152. The primary advantage of this approach is computational efficiency; while traditional models often require minutes or hours per training epoch, SkinNet-14 required only 2 to 8 seconds. This massive reduction in training time and resource usage suggests that Transformer-based models may represent the future of mobile-based skin cancer screening, enabling deployment on smartphones and low-end hardware in remote clinics without sacrificing diagnostic precision [3].

Summary of Literature

Collectively, these five studies illustrate a clear maturation in the field of automated skin cancer detection. The research has progressed from establishing CNNs as the baseline standard [5] to optimizing heavy transfer learning models for maximum sensitivity [4]. Currently, the frontier of research is defined by a dichotomy: one direction focuses on maximizing performance using complex attention mechanisms and segmentation [2], while the other focuses on extreme efficiency, proving that lightweight Custom CNNs [1] and Vision Transformers [3] can achieve state-of-the-art results even with limited data or low-resolution imagery. This study seeks to situate itself within this latter trajectory, investigating how optimized, efficient architectures can democratize access to life-saving diagnostic technology.

7. Research Design

Selected Design: Quantitative Research

Justification:

This study aims to empirically measure and compare the performance of computer algorithms. The nature of the data involved is inherently numerical and statistical, involving:

1. **Pixel Data:** The inputs are digital images represented as numerical matrices (224 X 224 X 3 or 32 X 32 X 3 arrays).
2. **Performance Metrics:** The evaluation relies on calculating precise statistical values including Accuracy (%), Sensitivity/Recall, Specificity, F1-Score, and Area Under the Curve (AUC).
3. **Computational Metrics:** We are measuring quantitative variables such as training time (in seconds/minutes), model parameters (in millions), and loss functions.

Qualitative methods (interviews or observations) are not suitable for measuring the mathematical efficiency or predictive power of a neural network. Therefore, a purely Quantitative experimental design is the only appropriate choice to test the hypothesis.

8. Research Variables

Independent Variables (IV):

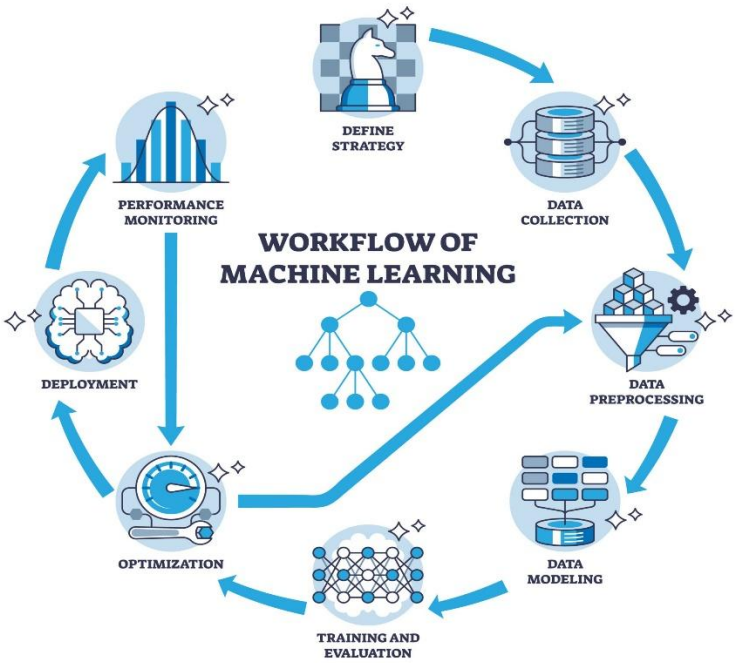
- **Model Architecture:** The specific type of Deep Learning algorithm used (e.g., Custom CNN, Compact Convolutional Transformer, DenseNet201, YOLOv7).
- **Image Resolution:** The dimensions of the input data (e.g., 224 X 224 pixels vs. 32 X 32 pixels).
- **Augmentation Strategy:** The techniques applied to the training data (e.g., Geometric augmentation vs. Photometric augmentation).

Dependent Variables (DV):

- **Classification Accuracy:** The percentage of correctly classified skin lesions.
- **Sensitivity (Recall):** The ability of the model to correctly identify positive cancer cases (crucial for medical diagnosis).
- **Training Time:** The duration required for the model to learn from the data (measured in seconds/epoch).
- **Model Complexity:** Measured by the number of trainable parameters and model size in megabytes.

9. Scientific Method

The research will follow a structured Deep Learning pipeline as utilized in the analyzed literature.



Step-by-Step Methodology:

Step 1: Data Collection utilizes two benchmark datasets HAM10000 ("Human Against Machine") and ISIC (International Skin Imaging Collaboration). These datasets were selected because they contain diverse dermoscopic images across seven diagnostic categories, including Melanoma (MEL), Melanocytic Nevi (NV), and Basal Cell Carcinoma (BCC). They also address common limitations noted in previous studies, such as poor diversity and severe class imbalance.

Step 2: Preprocessing focuses on resolving image quality issues. Raw images often contain artifacts like hair, ruler markings, and air bubbles, which reduce diagnostic accuracy. Therefore, morphological operations are used for artifact removal, followed by Non-Local Means Denoising (NLMD) and Contrast Limited Adaptive Histogram Equalization (CLAHE) to enhance contrast and lesion visibility. To support both traditional and lightweight model architectures, each image is resized into two resolutions: standard 224×224pixels for transfer learning models and low-resolution 32×32pixels for compact transformer-based designs.

Step 3: Data Augmentation is applied to address significant class imbalance, particularly the overrepresentation of Nevi images versus Melanoma. The pipeline incorporates geometric augmentations (rotation, horizontal and vertical flipping) and photometric adjustments (brightness, contrast, sharpness). To further strengthen minority classes, synthetic lesion samples generated using Generative Adversarial Networks (GANs) are also considered.

Step 4: Model Design and Configuration employs two architectures for comparison. Model A is DenseNet201, a widely used transfer-learning network known for high sensitivity and reduced false positives in melanoma detection. Model B is SkinNet-14, a compact Convolutional Transformer (CCT) variant that uses a tokenizer block and a single transformer encoder to efficiently process low-resolution images with reduced computational cost.

Step 5: Training and Validation uses a 70/10/20 split for the training, validation, and test subsets. Models are trained using the Adam optimizer with a learning rate of 0.001 and a batch size of 128, hyperparameters optimized through prior ablation studies. All experiments are conducted using deep learning frameworks (PyTorch/TensorFlow) accelerated on GPU hardware such as NVIDIA Tesla P100 or GTX 1660 to manage computational load.

Step 6: Performance Evaluation uses standard diagnostic metrics including Accuracy, Sensitivity (Recall), Specificity, and F1-Score. Efficiency metrics such as training time per epoch and parameter count are also recorded to assess computational trade-offs between DenseNet201 and SkinNet-14. A confusion matrix is generated to analyze misclassifications, with particular attention to visually similar classes such as BCC and SCC, providing insight into class-wise weaknesses and model reliability.

10. References

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